

Theme 5: Integration

Note Title

2014/04/20

Unit 5.1 Antiderivatives p 344 - 350

Definition: Function F is called an antiderivative of f on an interval if

$$F'(x) = f(x) \quad \forall x \in I$$

Example:

$$f(x) = x^2$$

$$F(x) = \frac{1}{3}x^3$$

$$F'(x) = x^2 = f(x)$$

$$f(x) = \sin x$$

$$F(x) = -\cos x$$

$$F'(x) = +\sin x = f(x)$$

But what about

$$f(x) = x^2$$

$$F(x) = \frac{1}{3}x^3 + 4$$

General antiderivative is

$$F(x) = \frac{1}{3}x^3 + C$$

Theorem 1 $\pm$ p 340

Examples:

Find the most general antiderivative of:

$$(1) \quad f(x) = x^4 \\ F(x) = \frac{1}{5}x^5 + C$$

$$f(x) = x^n \\ F(x) = \frac{x^{n+1}}{n+1} + C$$

$$(2) \quad f(x) = \sec^2 x \\ F(x) = \tan x + C$$

$$(3) \quad f(x) = \frac{1}{x} \\ F(x) = \ln|x| + C$$

$$(4) \quad f(x) = \operatorname{cosec} x \cot x \\ F(x) = -\operatorname{cosec} x + C$$

$$(5) \quad f(x) = \frac{1}{1+x^2} \\ F(x) = \arctan x + C$$

$$(6) \quad f(x) = \frac{1}{\sqrt{1-x^2}} \\ F(x) = \operatorname{arcsin} x + C$$

$$(7) \quad f(x) = \sec x \\ F(x) = ???$$

[WTW 168]

$$(8) \quad f(x) = 4 \sin x + \frac{2x^5 - \sqrt{x}}{x}$$

$$F(x) = -4 \cos x + \frac{2}{5} x^5 - 2x^{1/2}$$

$$\frac{2x^5 - \sqrt{x}}{x} = \frac{2x^5}{x} - \frac{\sqrt{x}}{x} = 2x^4 - x^{-1/2}$$

Function	Particular antiderivative	Function	Particular antiderivative
$c f(x)$	$c F(x)$	$\sec^2 x$	$\tan x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec x \tan x$	$\sec x$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
$\frac{1}{x}$ ✓	$\ln x $	$\frac{1}{1+x^2}$	$\tan^{-1} x$
e^x ✓	e^x	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$
$\sin x$	$-\cos x$		

Examples:

$$\textcircled{1} f(x) = 3e^x + 7\sec^2 x + \sec x \tan x.$$

$$F(x) = 3e^x + 7\tan x + \sec x + C$$

$$\textcircled{2} f(x) = \frac{2+x^2}{1+x^2} = \frac{1+x^2}{1+x^2} + \frac{1}{1+x^2}$$

$$= 1 + \frac{1}{1+x^2}$$

$$F(x) = x + \arctan x + \underline{\underline{C}} \rightarrow \text{Family of solutions.}$$

$$\textcircled{3} f'(x) = \frac{4}{\sqrt{1-x^2}} \quad \text{and} \quad f\left(\frac{1}{2}\right) = 1$$

$$f(x) = 4 \arcsin x + C$$

$$f\left(\frac{1}{2}\right) = 4 \arcsin\left(\frac{1}{2}\right) + C = 1$$

$$4\left(\frac{\pi}{6}\right) + C = 1$$

$$\Rightarrow C = 1 - \frac{2\pi}{3}$$

$$f(x) = 4 \arcsin x + 1 - \frac{2\pi}{3}$$

④ Find a antiderivative for

$$f(x) = e^{\sqrt{2x}} \sin(e^{2x})$$

$$F(x) = -\frac{1}{2} \cos(e^{2x})$$

Guess $\frac{d}{dx} \cos(e^{2x}) = -2 \cdot e^{2x} \sin(e^{2x})$

⑤ Find all the antider of

$$f(x) = \frac{\sec^2 \sqrt{1-x}}{\sqrt{1-x}}$$

Guess $\frac{d}{dx} \tan \sqrt{1-x}$

$$= \sec^2 \sqrt{1-x} \cdot \frac{1}{2} (1-x)^{-1/2} \cdot -1$$

$$F(x) = -2 \tan \sqrt{1-x} + C$$

⑥ Find the function f with

$$f'(x) = \frac{1}{3x+1} \text{ and } f(3) = 0$$

$$f(x) = \frac{1}{3} \ln|3x+1| + C$$

Guess: $\frac{d}{dx} \ln|3x+1| = \frac{3}{3x+1}$

$$f(3) = \frac{1}{3} \ln 10 + C = 0$$

$$C = -\frac{1}{3} \ln 10$$

$$f(x) = \frac{1}{3} \ln|3x+1| - \frac{1}{3} \ln 10$$

⑦ Find f if

$$f''(x) = \cos 2x \quad \text{and} \rightarrow \boxed{f'(0)=2 \text{ and } f(0)=-1}$$

$$f'(x) = \frac{1}{2} \sin 2x + C_1$$

$$f(x) = -\frac{1}{4} \cos 2x + C_1 x + C_2$$

$$\rightarrow f'(0) = C_1 = 2$$

$$\rightarrow f(0) = -\frac{1}{4} + C_2 = -1 \Rightarrow C_2 = -\frac{3}{4}$$

$$\Rightarrow f(x) = -\frac{1}{4} \cos 2x + 2x - \frac{3}{4}$$